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Signatures of the midnight open-closed magnetic field line boundary during balanced dayside and nightside reconnection

M. L. Parkinson, P. L. Dyson, M. Pinnock, J. C. Devlin, M. R. Hairston, E. Yizengaw, and P. J. Wilkinson

Abstract. The geomagnetic conditions were moderately disturbed ($K_p = 2$) during magnetic midnight on 10 December 1999, when the Tasman International Geospace Environment Radar (TIGER), a Southern Hemisphere HF SuperDARN radar, observed a persistent, sharp latitudinal decrease (~ 90 km) in spectral width near $-69^\circ\Lambda$. The line-of-sight Doppler velocity also rapidly declined across this spectral width boundary (SWB). The region poleward of the SWB was characterized by high spectral widths (>200 m/s), and the start of bursty equatorward and eastward flows (>500 m/s), which rapidly expanded equatorward. The relationships between familiar ionospheric and magnetospheric regions were inferred by comparing TIGER data with spectrograms calculated from precipitating particles measured on board the Defence Meteorology Satellite Program (DMSP) F14 satellite. The high spectral width scatter is often observed, and on this evening it was associated with irregularities forming on the open (but soon to be reconnected) field lines threading the polar cap ionosphere to the southern tail lobe. The region equatorward of the SWB was characterized by very low spectral widths (<50 m/s) and generally slower, more zonal flows (<300 m/s). This kind of scatter is more transient, and was associated with irregularities residing on the closed field lines threading the discrete and diffuse auroral oval to the plasma sheet boundary layer (PSBL) and central plasma sheet (CPS). Hence, the SWB was a reasonable proxy for the open-closed field line boundary, and the equatorward limit of the region, with low spectral width, was probably aligned with the poleward wall of the main ionospheric trough. The SWB was observed to contract poleward and expand equatorward on time scales of ~ 10 min, much as would be expected during balanced dayside and nightside reconnection. Total electron content (TEC) measurements made at Macquarie Island ($-65^\circ\Lambda$) and Hobart ($-54^\circ\Lambda$), and the ionograms recorded at the same stations, as well as at Bundoora ($-49^\circ\Lambda$), also helped to validate the interpretation.

Key words. Ionosphere (auroral ionosphere; electric fields and currents; ionosphere-magnetosphere interactions)